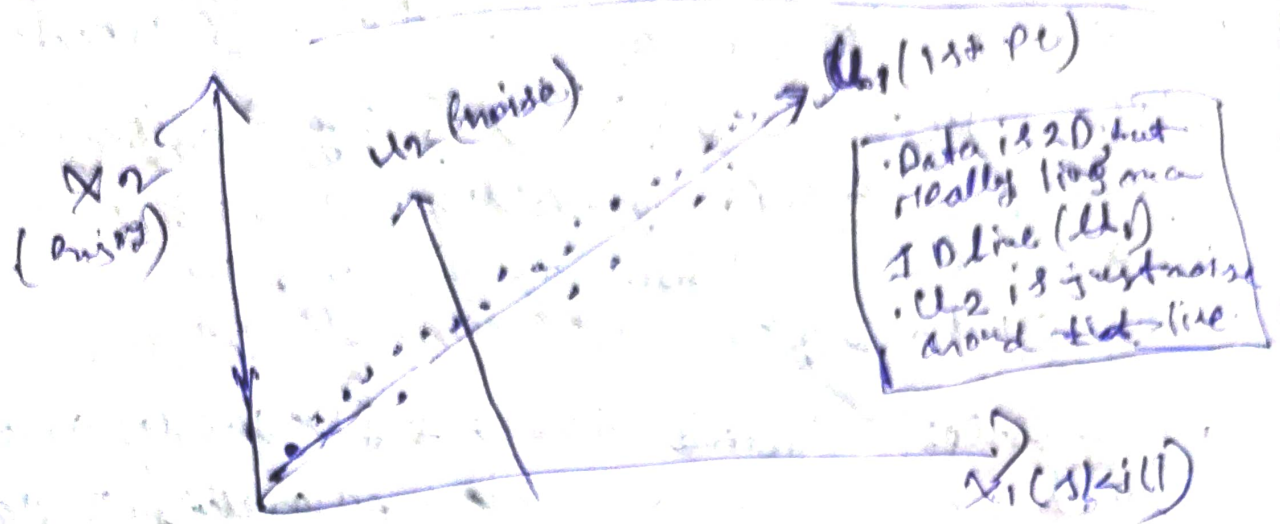


Principle Component Analysis



The core Idea of PCA:-

your data lives in n dim, but doesn't actually use all of them. Most real-world data clusters along a lower-dim structure - like those pilot points above. They are technically 2D (skill + ensight), but they fall almost perfectly on a diagonal line. That diagonal is the signal. Everything perpendicular to it is noise.

(PCA finds these signal directions automatically).

The key insight: variance = information. When we project data onto a direction, the resulting spread (variance) tells us how much information that direction holds.

Project onto u_1 (diagonal): points spread out wide $\rightarrow$ high variance $\rightarrow$ lots of info retained.

• project onto u_2 (perpendicular): points cluster near zero $\rightarrow$ low var $\rightarrow$ almost no info.
 So: find the direction of max var = find the 1st principal component.

• The math (es) $\rightarrow$

St-1: Pre-process

- zero the mean: subtract average from each point.
- Normalize variance: divide each feature by its std dev (so mph & km/h don't fight each other)

St-2: Find the direction 'u' that maximizes variance of projections.

the projection of point x onto unit vector 'u' is: $x^T u$ (a scalar)

Variance of all projections:

$$\frac{1}{n} \sum (x^{(i)T} u)^2 = u^T \Sigma u$$

where $\Sigma = \frac{1}{n} \sum x^{(i)} x^{(i)T}$ is the empirical covariance matrix.

St-3: maximize $u^T \Sigma u$ subject to $\|u\|=1$

soln: u = principal eigenvector of Σ (eigenvector with largest eigenvalue).

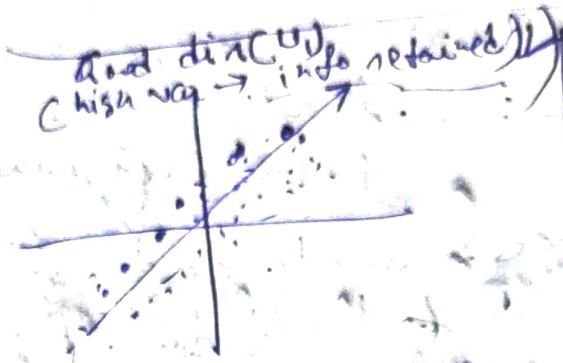
Why? Because $\Sigma u = \lambda u$, so $u^T \Sigma u = \lambda u^T u = \lambda$

maximum λ = maximum variance.

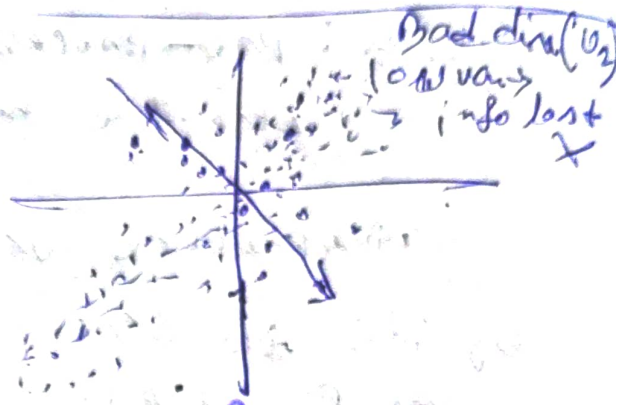
St 4: For k dimensions, take top k eigen-vectors.
 Your compressed representation of $X^{(i)}$:

$$y^{(i)} = [u_1^T x^{(i)}, u_2^T x^{(i)}, \dots, u_k^T x^{(i)}]^T \in \mathbb{R}^k$$

This is an $n \rightarrow k$ dimensionality reduction.



projections spread apart (high variance)



Projections clump together

(compression 16x18 to 6x6)

Full PCA Algorithm:

1. compute mean μ , subtract it from all points.
2. Normalize by std dev per feature.
3. Compute covariance matrix $Z = (1/n) X^T X$
4. Find eigenvectors of Z , sorted by Eigenvalue (largest first)
5. Pick top k eigenvectors $\rightarrow$ these are $u_1, \dots, u_k$ principal components.
6. Project: $y^{(i)} = [u_1^T x^{(i)}, \dots, u_k^T x^{(i)}]^T$

Why eigen-vector, Eigenvalues, Covariance matrix $\rightarrow$

Concept	Role in P.C.A
Covariance matrix Σ	Encodes all pairwise relationships in our data
Eigenvectors of Σ	The principal directions (axes of the data cloud)
Eigenvalues of Σ	How much variance each direction holds

